

Panini Linguistics Olympiad 2023

Round 1 (Junior), 5th March 2023
Question Booklet

Maximum marks: 100

Total time: 4 hours

- This question booklet has 8 pages, including this page. When the exam begins, please check and inform the invigilator immediately if your booklet has less than 8 pages.
- This question booklet has 5 problems. Each problem can have multiple assignments; the marks for each assignment are indicated next to the assignment, and the total marks for the problem are indicated next to the problem title.
- **It is not expected that you will solve every problem.** It is recommended to read all the problems before starting to solve any. You don't need to solve them in order.
- At the end of some problems, we've given you some extra information and interesting facts related to the problems in footnotes. They are not required to solve the problems but may sometimes be helpful.
- All problems will ask for short objective answers (e.g. match the following, multiple choice, one-line answers, etc.). Some problems may also ask for detailed explanations of your solutions. Your answer booklet has space for both.
 - Please write all your answers in the **specific space provided** in the answer booklet.
 - Space has also been provided for answers that require explanations. If you need extra space, you may continue outside the given space, or on the back of the sheet, or on a blank sheet that the invigilator will give you. You don't need to repeat the other answers in your explanations but you can refer to them by their number if you want to.
 - You don't need to copy the questions, problem statements, or any of the data in your answers, but you can refer to them in your own way as long as it is clear.
 - Please write the **problem number**, your **roll number**, and your **name** on **every sheet** of the answer booklet, including any extra sheets.
- You are not allowed to use any calculators, mobile phones, other electronic devices, books, notes, or other external sources of information during the exam. If you have any questions at any time, you should ask the invigilator, but you are not allowed to communicate with anyone else during the exam.

All the best!

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Problem 1: Nanti

(20 marks)

Here are some constructions in Nanti.¹

igemarite	his tapir
nobonkiti	my foot
iserine	his tobacco
oshiropitonkite	her needle
nogusine	my piping guan
pirirana	your blood
nobarotote	my raft
pibirane	your flashlight
pigotsirote	your knife
igabeharite	his bird-trap
ogito	her head
ibanko	his house
ogobitite	her pot
nogomaginarote	my woolly monkey
ogetane	her walnut
pitsitsihenkate	your soot

Assignment 1: (6 marks)

Translate from Nanti:

- | | |
|-------------|----------------|
| (a) kemari | (d) pibanko |
| (b) irirana | (e) ogotsirote |
| (c) ponkiti | (f) noserine |

Given below are some additional Nanti nouns.²

kuri	peach palm
pori	leg
saburi	machete

¹The Nanti language is spoken by ~250 people in Peru. The tapir is a large, pig-like animal. The piping guan is a South American bird species.

²The peach palm is a species of palm tree native to Central and South America.

Assignment 2: (6 marks)

Translate to Nanti:

- | | |
|-------------------|---------------------|
| (a) your leg | (d) raft |
| (b) my peach-palm | (e) woolly monkey |
| (c) his machete | (f) her piping guan |

Assignment 3: (8 marks)

Explain your solution.

– Antara Raaghavi Bhattacharya

Problem 2: Romanian

(25 marks)

Here are some phrases in Romanian³ and their English translations:⁴⁵

copilul micuț al cetățeanului viteaz	the small child _M of the brave citizen _M
aptul cetățean al nepoatei moarte	the skilled citizen _M of the dead niece
camarazii curioși ai sătulelor cetățene	the curious comrades _M of the satiated citizens _F
viclenii nepoți ai înalților copii	the cunning nephews of the tall children _M
camarada smeadă a mireselor flămânde	the pale comrade _F of the hungry brides
nepoata vicleană a curiosului urs	the cunning niece of the curious bear _M
frumosul mânz al urșilor apți	the beautiful foal _M of the skilled bears _M
goalele nomade ale micuței camarade	the underdressed nomads _F of the small comrade _F
profesorul viteaz al hoților smezi	the brave teacher _M of the pale thieves _M
sătulii studenți ai hoatei înalte	the satiated students _M of the tall thief _F
mireasa înaltă a flămânzilor profesori	the tall bride of the hungry teachers _M
moartele verișoare ale frumoaselor profesoare	the dead cousins _F of the beautiful teachers _F

Assignment 1: (6 marks)

Translate to English:

- (a) hoțul flămând al copilelor viteze
- (b) micuța cetățeană a mânjilor frumoși
- (c) studentele viclene ale înaltului nepot

³Romanian belongs to the Romance branch of the Indo-European family. It is spoken by approx. 24 million people primarily in Romania and Moldova.

⁴ș = sh in shop; ț = ts in cats; j = s in vision; ă = a in about; â and î are vowels.

_M = masculine; _F = feminine.

⁵A foal is a baby horse.

Assignment 2: (6 marks)

Translate to Romanian. If multiple translations are possible, write only one:

- (a) the brave cousins_M of the dead foals_M
- (b) the satiated thief_F of the underdressed nomads_M
- (c) the curious cousin_F of the skilled child_F

Assignment 3: (4 marks)

Translate into Romanian in all possible ways:

- (a) the small citizens_M of the underdressed nomad_M

Assignment 4: (9 marks)

Explain how these phrases are structured in Romanian.

– Vlad A. Neaşcu

Problem 3: Icari Dargwa

(15 marks)

Here are some sentences in Icari Dargwa⁶ and their English translations in random order:⁷

- | | |
|------------------------|---|
| A. dul mazi ucibda | i. I caught the cat. |
| B. nusal tati ucibda | ii. The chief caught me. |
| C. nusal tati ucibdi | iii. The chief caught the cat. |
| D. tatil mazi čidažib | iv. The father saw the cat. |
| E. ul kurta čidažibdi | v. The girl saw the horse. |
| F. ustal du ucibda | vi. We caught the father. |
| G. ustal mazi ucib | vii. You _{pl} caught the father. |
| H. rirsil jabu čidažib | viii. You _{sg} saw the fox. |

Assignment 1: (4 marks)

Determine the correct correspondences.

Assignment 2: (2 marks)

Translate into English:

- (a) rirsil nusa ucibdi
- (b) mazil du čidažibda

⁶Icari Dargwa is a variant of the Dargwa language, which belongs to the Northeast Caucasian (Nakh-Dagestani) family. It is spoken by approx. 497 thousand people in the Republic of Dagestan, Russia.

⁷c = ts in tsar, č = ch in chop, ž = s in vision.

Assignment 3: (6 marks)

Translate into Icarian Dargwa:

- (a) The father caught the girl.
- (b) The horse caught me.
- (c) The fox saw us.

Assignment 4: (3 marks)

Explain your solution.

– Kazune Sato

Problem 4: Epun

(22 marks)

Here are some sentences in Epun⁸ and their English translations:⁹

Fa midebiniz kakolus fa vakvepiz.
Chavetius fa koviniz gub afa midep.
Asa hasmingus afa chegodiniz.
Guza hakakolgu afa zenadin.
Chupanibu fa pelikin.
Osozeu uza afa kop.
Afa bozudin cherehelgu fa vletop.
Nisa osozenis fa chegodopiz.
Afa kopiz afa bozudiniz havetigus.
Fa zaddiliniz hapanibus.

The kings see the walruses_M.
The children_M won't love the queen.
The dogs_F haven't kissed them.
The girl hasn't seen her.
The turtle_M won't come.
He doesn't catch the child_F.
The teacher_F will write the letter.
We catch the dogs_M.
The teachers_F have loved the children_F.
The men have come.

Assignment 1: (6 marks)

Translate into English:

- (a) Uza hapanibu.
- (b) Uza havetigu guza.
- (c) Fa pelikop nisa hakakolnis.

Assignment 2: (8 marks)

Translate into Epun:

- (a) The walrus_F will see us.

⁸Epun is an invented language, deliberately devised to contain constructions which violated universal grammatical principles.

⁹ch = ch in chop. The subscripts M and F refer to masculine and feminine, respectively.

- (b) The man hasn't written the letters.
- (c) The turtles_F don't catch the woman.
- (d) The children_M have loved the queens.

Assignment 3: (8 marks)

Explain how these sentences are structured in Epun.

– Vlad A. Neacșu (inspired by a problem created by Adina Dragomirescu)

Problem 5: Maldivian

(18 marks)

Below are four lines of a poem written in the Maldivian¹⁰ language, using the Thaana writing system. The Latin transcriptions in scrambled order are given on the right.¹¹

Some words are missing from the Latin transcriptions and have been replaced by placeholders such as [1], [2], etc.

- | | |
|----------------------------------|-------------------------------------|
| A. ޅަނަ ޅަނަ ޅަނަ ޅަނަ ޅަނަ ޅަނަ | i. ekam [1] avaha eythi geleyne |
| B. ޅަނަ ޅަނަ ޅަނަ ޅަނަ ޅަނަ ޅަނަ | ii. [2] mage [3] reethi lhen |
| C. ޅަނަ ޅަނަ ޅަނަ ޅަނަ ޅަނަ ޅަނަ | iii. moonu balaa loagadun [4] soora |
| D. ޅަނަ ޅަނަ ޅަނަ ޅަނަ ޅަނަ ޅަނަ | iv. thi [5] enme fahu loabi |

Assignment 1: (4 marks)

Determine the correct correspondences between lines (A-D) written in Thaana and the transcriptions (i-iv).

Assignment 2: (2 marks)

One of the words in Maldivian is written in an unexpected way. What is this word? Explain your answer.

Assignment 3: (5 marks)

Fill in the blanks [1] to [5].

Assignment 4: (3 marks)

Write the following names of islands in Thaana:

¹⁰Maldivian (also known as Dhivehi) is an Indic language spoken by approximately 350,000 people in the Maldives, and is the official language.

¹¹lh and th are consonants.

- (a) finey
- (b) baarah
- (c) angaagau

Assignment 5: (4 marks)

Explain how the Thaana writing system works. You do not need to list the meaning of every symbol.

– Vlad A. Neacșu

This is the end.